

Approximation Algorithm for Connected Roman k -Dominating Set

Mengmeng He ^{*}, Ralf Klasing [†], Yaping Mao [‡], Xiaoyan Zhang [§]

Abstract

Let k ($k \geq 2$) be a positive integer, and let G be a simple graph with the vertex set $V(G)$. A Roman k -dominating function (Rk-DF) on G is a function $f_{kR} : V(G) \rightarrow \{0, 1, 2\}$ such that every vertex u with $f_{kR}(u) = 0$ is adjacent to at least k vertices $v_1, v_2, \dots, v_k$ with $f_{kR}(v_i) = 2$ for $i = 1, 2, \dots, k$. The minimum Roman k -dominating set problem aims to compute a Roman k -dominating function f_{kR} that minimizes the total weight $\sum_{v \in V} f_{kR}(v)$. The minimum Connected Roman k -dominating set problem (MinCRkDS) seeks to find a minimum-weight Roman k -dominating function f_{kR} such that the subgraph of G induced by $D_{kR} = \{v \in V \mid f_{kR}(v) = 1 \text{ or } f_{kR}(v) = 2\}$ is connected. As far as we know, this paper is the first paper to solve MinCRkDS in general graphs. We present a greedy algorithm for MinCRkDS with an approximation ratio $(1 + \varepsilon)(2 + \ln(k + 1 + 2\Delta))$ for any $\varepsilon > 0$, where Δ is the maximum degree of the graph.

Key words: Roman k -dominating set · Connected Roman k -dominating set · Greedy algorithm

· Non-submodular · Approximation ratio

1 Introduction

The concept of the *Roman dominating function* originates from an article titled “Defend the Roman Empire!” published by Stewart in 1999 [16]. As per [1], in the 4th century A.D., Constantine the Great, the Emperor of Rome, issued a decree for the defense of his cities. The decree stated that any city lacking a stationed legion for its protection must be adjacent to another city with two stationed legions. In the event of an attack on the first city, the second city could deploy a legion to safeguard it without exposing itself to vulnerability. The primary goal was to minimize the overall number of legions required.

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However, given the numerous adversaries the Roman Empire faced, if a set of k enemies attacked k cities without a legion, these cities could be deemed secure if they were adjacent to at least k cities, each with two stationed legions. This naturally leads to the generalization of the *Roman k -Dominating Function*.

A *Roman k -dominating function* (Rk-DF) on G is a function $f_{kR} : V(G) \rightarrow \{0, 1, 2\}$ such that each vertex u with $f_{kR}(u) = 0$ is connected to at least k vertices $v_1, v_2, \dots, v_k$ with $f_{kR}(v_i) = 2$ for $i = 1, 2, \dots, k$. Let $V_{kR}^0, V_{kR}^1, V_{kR}^2$ denote the sets of vertices with values 0, 1, 2 under the function f_{kR} , respectively. The *weight* of a Roman k -dominating function is the value $\sum_{v \in V} f_{kR}(v)$. The minimum weight of a Roman k -dominating function on a graph G is termed the *Roman k -domination number* $\gamma_{kR}(G)$ of G . It is noteworthy that the Roman 1-domination number $\gamma_{1R}(G)$ corresponds to the conventional *Roman domination number* $\gamma_R(G)$.

The definition of the *Roman k -dominating function* can also be inspired by the server placement described below. When a request can occur at any node in a graph and requires a server at that node, a *minimum dominating set* represents a minimum set of servers that serve an arbitrary single request by moving a server along at most one edge. The *minimum Roman dominating set* represents the minimum number of servers to place when there are two requests to access different nodes, requiring each vertex to either possess its own server or to have a neighbor with two servers. If there are k different requests accessing k different vertices at the same time, in this case, how to place the server to ensure that each request has a corresponding server response? Hence, how to find a method of placing the servers corresponds to how to find an *Roman k -dominating function*, and computing a most effective placement of servers corresponds to computing a minimum *Roman k -dominating function*.

The *minimum Roman k -dominating set problem* involves determining an Roman k -dominating function f_{kR} that minimizes the weight $\sum_{v \in V} f_{kR}(v)$. In specific scenarios, the requirement is for the various legions to communicate with each other, and this communication is ensured through connectivity. This concept leads to the *minimum connected Roman k -dominating set problem* (MinCR k DS), whose objective is to compute a minimum Roman k -dominating function f_{kR} such that the subgraph of G induced by $V_{kR}^1 \cup V_{kR}^2$ is connected.

In a recent study, Mohanapriya et al. [17] delved into the computational complexity of identifying a minimum Roman k -dominating function (Rk-DF) on split graphs. Their research revealed that determining minimum Rk-DF's on star-convex bipartite graphs and comb-convex bipartite graphs are NP-complete problems. Additionally, they demonstrated that the task of finding minimum Rk-DF on bipartite chain graphs is solvable in polynomial time. Addressing the approximation perspective, they established that there exists no constant factor approximation algorithm for minimum Rk-DF.

The minimum weight of an Roman k -dominating function on a graph G is termed the *Roman k -domination number* $\gamma_{kR}(G)$. Hansberg et al. [6], employing the probabilistic method, established new bounds for the Roman k -domination number $\gamma_{kR}(G)$. Padamutham et al. proposed a straightforward approach for computing an Roman dominating set in [13]. The method involves initially calculating a dominating set D using a $(1 + (\ln(\Delta + 1)))$ -approximation algorithm. Subsequently, assigning the value 2 to every vertex in D and the value 0 to the remaining vertices

increases the approximation ratio by a factor of 2. Similarly, by utilizing an existing effective approximation algorithm, we can compute the fault-tolerant dominating set (it requires that each vertex v in the graph either belongs to the dominating set or is adjacent to at least k vertices in the dominating set). First, compute a fault-tolerant dominating set D and then assign the value 2 to every vertex in D and the value 0 to the other vertices. Consequently, this provides an approximation algorithm for *the minimum Roman k -dominating set problem*. For recent algorithmic studies addressing variations of the Roman domination problem, please refer to [8, 10, 14, 20, 11].

The study in [12] demonstrated that the decision problem associated with computing γ_{kR} is NP-complete, even when constrained to bipartite and chordal graphs. Additionally, for the case of $k = 2$, the decision problem remains NP-complete even in the context of planar graphs. Shang et al. [18] showed that the Roman 1-domination problem in unit disk graphs is NP-hard. And then presented a simple linear time approximation algorithm and a polynomial-time approximation scheme for this problem, respectively. Haynes et al. [7] established that a linear algorithm for computing the Roman k -domination number for any tree has not yet been developed. To the best of our knowledge, there is currently no existing approximation algorithm for MinCR k DS in general graphs. This paper addresses this gap by introducing a $(1 + \varepsilon)(2 + \ln(k + 1 + 2\Delta))$ approximation algorithm for MinCR k DS in general graphs.

1.1 Our Contributions

In this paper, we introduce a novel greedy algorithm to address the *minimum connected Roman k -dominating set problem*. The algorithm achieves an approximation ratio of $(1 + \varepsilon)(2 + \ln(k + 1 + 2\Delta))$, where $\varepsilon > 0$ is an arbitrary constant. Notably, in each iteration of the greedy algorithm, a constant number of vertices are added to the currently selected subset of vertices. It is important to clarify that the "add" operation does not constitute a union in the conventional sense; rather, we allow each vertex to be selected at most twice. A vertex selected $t \in \{0, 1, 2\}$ times is assigned an f_{kR} -value of t . This f_{kR} -value assigned to a vertex can be interpreted as its cost within the cost-effectiveness of the greedy selection. Unlike the classic greedy algorithm, this cost is not predetermined but dynamically changes, posing a significant challenge for our analysis algorithm.

We know that potential functions with submodular properties play an important role in the analysis of approximation ratios of greedy algorithms, but another challenge in this paper lies in the non-submodular nature of the potential function we construct. Furthermore, the analysis of the algorithm encounters the difficulty of establishing a crucial inequality (5), which plays an important role in the process of analyzing the approximation ratio of the greedy algorithm. Despite these challenges, our objective remains obtaining the desired approximation ratio. To achieve this, a key step involves establishing a connection between the optimal solution and the solution derived from our algorithm through a suitable decomposition of the optimal solution. This marks the first-ever approximation algorithm for MinCR k DS in general graphs.

The remainder of this paper is structured as follows. In Section 2, we introduce some of the symbols and preparatory knowledge used in this article. In Section 3, we begin by constructing the potential function, subsequently delving into its properties, presenting the algorithm, and

conducting a theoretical analysis. Section 4 serves as the conclusion, summarizing the key findings, and offering insights into potential avenues for future research.

2 Preliminaries

Without loss of generality, we assume that all graphs discussed in this paper are connected. Let $G = (V, E)$ be a graph with a vertex set $V = V(G)$ and an edge set $E = E(G)$. For a vertex subset $C \subseteq V$, we denote by $N_C(v)$ the set of vertices in C that are adjacent to v in G for every vertex $v \in V$. Specifically, we use $N_G(v)$, referred to as the *open neighborhood* of a vertex $v \in V$, to represent the set of nodes adjacent to v in G . The *closed neighborhood* of a vertex $v \in V$ is defined as the set $N_G[v] = N_G(v) \cup \{v\}$. The *open neighborhood* of a set $C \subseteq V$ is expressed as $N_G(C) = \cup_{v \in C} N_G(v)$. Similarly, the *closed neighborhood* of a set $C \subseteq V$ is given by $N_G[C] = N_G(C) \cup C$. The subgraph of G induced by the vertex set C , denoted as $G[C]$, is the graph with the vertex set $V(G[C]) = C$ and the edge set $E(G[C]) = \{uv : u, v \in C \text{ and } uv \in E(G)\}$. A vertex set is considered connected if it induces a connected subgraph.

Definition 2.1. [3] Consider a finite set E and a set function $f : 2^E \rightarrow \mathbb{R}^+$, where 2^E denotes the power set of E —the collection of all subsets of E . The function f is termed *submodular* if, for any two elements A and B in 2^E ,

$$f(A) + f(B) \geq f(A \cap B) + f(A \cup B).$$

This inequality characterizes the submodularity property of the function f .

Definition 2.2. [3] Assume that $f(\cdot)$ is a submodular set function on subsets of E . Define

$$\Delta_B f(A) \equiv f(A \cup B) - f(A)$$

for any subsets A and B of E ; that is, $\Delta_B f(A)$ is the extra amount of $f(\cdot)$ value we gain by adding B to A . When $B = \{v\}$ is a singleton, we simply write

$$\Delta_v f(A) = f(A \cup \{v\}) - f(A) \tag{1}$$

instead of $\Delta_{\{v\}} f(A)$.

In this article, each $C \subseteq V$ is partitioned into two disjoint parts: $C = C^1 \cup C^2$, where C^i ($i = 1, 2$) represents the set of vertices with f_{kR} -value = i . For two sets $C = C^1 \cup C^2$ and $D = D^1 \cup D^2$, we declare that $C \subseteq D$ if $C^1 \subseteq D^1$ and $C^2 \subseteq D^2$. The union of C and D is $C \cup D = (C \cup D)^1 \cup (C \cup D)^2$, where we initially set $(C \cup D)^t = C^t \cup D^t$ for $t = 1, 2$. Subsequently, for every vertex v with $v \in (C \cup D)^1 \cap (C \cup D)^2$, we remove v from $(C \cup D)^1$. This union operation can be viewed as a prioritized union, giving precedence to the higher f_{kR} -value. The intersection of sets C and D is $(C \cap D) = (C \cap D)^1 \cup (C \cap D)^2$, where $(C \cap D)^t = C^t \cap D^t$ for $t = 1, 2$. The subsequent example illustrates these union and intersection operations (see Figure 1).

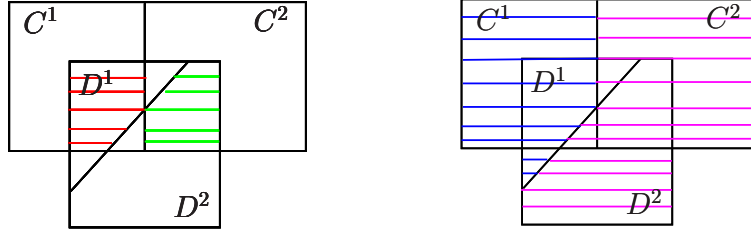


Figure 1: The left side of the diagram illustrates the red-shaded portion, representing $(C \cap D)^1$, while the green-shaded area depicts $(C \cap D)^2$. On the right side, the blue-shaded region represents $(C \cup D)^1$, and the purple-shaded area represents $(C \cup D)^2$.

3 Greedy Algorithm for the MinCR k DS

3.1 Potential Function

For a connected graph $G = (V, E)$ and a vertex subset $C = C^1 \cup C^2 \subseteq V$, a function f_{kR} associated with C is define to be

$$f_{kR}^C(v) = \begin{cases} 1, & v^1 \in C^1, \\ 2, & v^2 \in C^2, \\ 0, & v^1 \notin C^1 \text{ and } v^2 \notin C^2. \end{cases} \quad (2)$$

Define

$$S_i^{C^2} = \{v \in V \setminus C : |N_{C^2}(v)| = i\} \quad (i = 0, 1, \dots, k-1)$$

and

$$S_k^{C^2} = \{v \in V \setminus C : |N_{C^2}(v)| \geq k\}.$$

We say that a vertex v is *black* if $v \in C^2$, *gray* if $v \in C^1$, *red* if $v \in S_k^{C^2}$, *pink* if $v \in (\cup_{i=1}^{k-1} S_i^{C^2})$, and *white* if $v \in S_0^C$. Denote by $\mathcal{B}_C$, $\mathcal{G}_C$, $\mathcal{R}_C$, $\mathcal{P}_C$, and $\mathcal{W}_C$ the set of black, gray, red, pink, and white nodes with respect to C , respectively.

We define the potential function g as follows. For a vertex subset $C \subseteq V$, define

$$r(C) = \sum_{v \in V} r_C(v), \quad (3)$$

where

$$r_C(v) = \begin{cases} 0, & v \in C^1 \text{ or } v \in C^2 \text{ or } v \in S_k^{C^2}, \\ k - i, & v \in (\cup_{i=1}^{k-1} S_i^{C^2}), \\ k, & v \in S_0^C. \end{cases} \quad (4)$$

Define $p(C)$ as the count of connected components in the subgraph $G[C]$ induced by C . Let $E_N(C)$ represent the set of edges with one end in $N_G(C^2) \setminus C$ and the other end in C . We utilize $q(C)$ to signify the number of connected components in the graph $G \langle C \rangle$ with the vertex set V and the edge set $E(G[C]) \cup E_N(C)$.

Let us define the potential function g as

$$g(C) = r(C) + p(C) + q(C).$$

Lemma 1. *For vertex subset $C \subseteq V$, C is a connected Roman k -dominating set if and only if $g(C) = 2$.*

Proof. Suppose C is a connected Roman k -dominating set. Consequently, $G[C]$ is connected, indicating $p(C) = 1$. Moreover, every vertex $v \in V \setminus C$ is adjacent to at least k vertices in $C^2 \subseteq V(G[C])$, implying $v \in S_k^{C^2}$ and ensuring $G \setminus C$ is connected (thus $q(C) = 1$). Utilizing (3) and (4), we observe $r(C) = \sum_{v \in V} r_C(v) = \sum_{v \in V} 0 = 0$. Consequently, $g(C) = r(C) + p(C) + q(C) = 2$. The necessity has been demonstrated.

Conversely, assume $g(C) = 2$. We assert that C is a connected Roman k -dominating set through the following three claims.

Claim 1. $C \neq \emptyset$.

Proof. Assume $C = \emptyset$ for a contradiction. By the definitions of $r(\cdot), p(\cdot), q(\cdot)$, we have $r(\emptyset) = \sum_{v \in V} r_{\emptyset}(v) = k|V|$, $p(\emptyset) = 0$, $q(\emptyset) = |V|$, and $g(\emptyset) = r(\emptyset) + p(\emptyset) + q(\emptyset) = (k+1)|V|$. Since $|V| \geq 1$ and $k \geq 2$, then $g(\emptyset) = (k+1)|V| \geq 3$. This contradiction establishes Claim 1. $\square$

Claim 2. *For any vertex $v \in V$, $r_C(v) = 0$.*

Proof. Suppose there exists a vertex $v_0 \in V$ such that $r_C(v_0) > 0$ for a contradiction. According to the definition of $r_C(\cdot)$ and node shading, v_0 is either a pink vertex or a white vertex. If v_0 is a white vertex, then $r_C(v_0) = k \geq 2$. If v_0 is a pink vertex, then $r_C(v_0) = k - i$ ($i = 1, 2, \dots, k-1$) ≥ 1 . Since $C \neq \emptyset$, it follows that $p(C) \geq 1$ and $q(C) \geq 1$. Therefore, $g(C) = r(C) + p(C) + q(C) \geq 3$, which contradicts $g(C) = 2$. Claim 2 holds. $\square$

By the definition of $r(\cdot)$ and Claim 2, we conclude that C is a Roman k -dominating set.

Claim 3. $G[C]$ is a connected graph.

Proof. Combining Claims 1 and 2, it follows that $r(C) = 0$, $p(C) \geq 1$, $q(C) \geq 1$. Therefore, $g(C) = 2$ if and only if $p(C) = q(C) = 1$, implying that $G[C]$ is connected. $\square$

From these three claims, C is a connected Roman k -dominating set. $\square$

3.2 Approximation Algorithm

In this section, we introduce a greedy algorithm named Algorithm 1 for MinCR k DS. For a MinCR k DS instance $G = (V, E)$, we create two copies of all vertices $v \in V$ and designate them as v^1 and v^2 . Each vertex v is treated as copies v^1 and v^2 with costs 1 and 2, respectively. One of the challenges in solving the MinCR k DS is that choosing v^1 may lead to the possibility of selecting v^2 . To address this issue, we present an additional greedy algorithm.

Our algorithm permits each vertex in V to be selected at most twice. In line 3 of Algorithm 1, the multiset X may contain t copies of vertex v , where $t \in \{0, 1, 2\}$, and the contribution of vertex v to $|X|$ is t . For example, if the algorithm chooses $X = \{v_1, v_2, v_3\}$ in some iteration, then $|X| = 3$. It is important to note that every vertex can be selected at most twice. If a vertex v has already been selected once, then v may be chosen at most one more time in the subsequent X ; if v has been selected twice, then v will not be chosen again in the subsequent process. For a set $C = C^1 \cup C^2$ and a multiset X , define the new set (not multiset)

$$(C + X) = (C + X)^1 \cup (C + X)^2,$$

where $(C + X)^1 = \{v | v \in C^1 \text{ and } v \notin X\} \cup \{v | v \notin C \text{ and } v \text{ occurs once in } X\}$, $(C + X)^2 = \{v | v \in C \text{ or } v \in X\} \setminus (C + X)^1$. (see Figure 2)

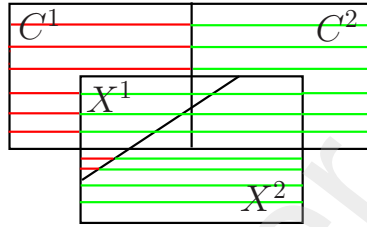


Figure 2: The red shaded part represents $(C + X)^1$, and the green shaded part represents $(C + X)^2$.

In other words, if $v \notin C$, add v to C^t if v is selected t times by X ; if v is already in C^1 and is selected once more by X , move v to C^2 note that if $v \in C^2$, then v cannot be selected by X . For two subsets $A, B \subseteq V$, we define the variant marginal profit of A over B as

$$\Delta_{+A}g(B) = g(B + A) - g(B).$$

The algorithm includes a parameter m related to the error term ε , which will be determined later. In line 3 of Algorithm 1, a multiset X with $|X| \leq 2m - 1$ that maximizes $\frac{g(C) - g(C+X)}{|X|}$ is chosen. The update of C^1 and C^2 is in the process of defining $C + X$. The update of f_{kR} follows (2).

Algorithm 1 Greedy Algorithm for MinCRkDS

Input: A connected graph $G = (V, E)$ and a positive integer $m \geq 2$.

Output: A connected Roman k -dominating function f_{kR} for G .

1: Set $C^1, C^2 \leftarrow \emptyset, C \leftarrow C^1 \cup C^2$;

2: **while** $g(C) > 2$ **do**

3: Select a set $X \subseteq V$ of size $|X| \leq 2m - 1$ that maximizes $-\frac{\Delta_{+X}g(C)}{|X|}$;

4: $C \leftarrow C + X$

5: Update C^1, C^2 and f_{kR}

6: **end while**

7: **return** f_{kR}

Lemma 2. Suppose $G = (V, E)$ is a connected graph with at least three vertices, and $C = C^1 \cup C^2$ is a vertex set such that $g(C) > 2$. Then, there exists a vertex $v \in V \setminus C$ such that $g(C + v) < g(C)$ or $g(C + v, v) < g(C)$.

Proof. Lemma 1 indicates that C is not a connected Roman k -dominating set. Therefore, there are two scenarios: either C is not a Roman k -dominating set, or $G[C]$ is not connected.

Case 1. C is not an Roman k -dominating set.

In this case, $C \cup S_k^{C^2} \neq V$. That means there are pink or white vertices.

Subcase 1.1. v is a pink vertex.

Clearly, $v \in \cup_{i=1}^{k-1} S_i^{C^2}$, and $r_C(v) = k - i$ ($i = 1, 2, \dots, k - 1$). Examine that $\Delta_{+\{v,v\}}g(C) = \Delta_{+\{v,v\}}r(C) + \Delta_{+\{v,v\}}p(C) + \Delta_{+\{v,v\}}q(C)$. Now, let's consider these three increments separately. For $\Delta_{+\{v,v\}}r(C)$, we have

$$\begin{aligned}\Delta_{+\{v,v\}}r(C) &= \sum_{u \in V} (r_{C+\{v,v\}}(u) - r_C(u)) \\ &= -r_C(v) - |N(v) \cap (\cup_{i=1}^{k-1} S_i^{C^2})| - |N(v) \cap S_0^{C^2}| \\ &\leq -(k - i) \leq -1 < 0,\end{aligned}$$

hence $r(C + \{v, v\}) < r(C)$. For $\Delta_{+\{v,v\}}p(C)$, we have

$$\Delta_{+\{v,v\}}p(C) = p(C + \{v, v\}) - p(C) = p(C \cup \{v^2\}) - p(C).$$

Considering $v \in \cup_{i=1}^{k-1} S_i^{C^2}$, it follows that v has at least one adjacent vertex in C^2 , placing v on some connected component of $G[C]$. If v^2 is added to C^2 , the number of connected components of $G[C]$ remains the same, so $\Delta_{+\{v,v\}}p(C) \leq 0$.

Now, let us analyze $\Delta_{+\{v,v\}}q(C)$. According to the definition of $q(\cdot)$, $q(C)$ represents the number of connected components in $G\langle C \rangle$. As v is a pink vertex, it is part of a connected component in the graph $G\langle C \rangle$. If v is adjacent to the remaining white vertices, including v in C^2 reduces the number of white connected components adjacent to v . In this case, $\Delta_{+\{v,v\}}q(C) = q(C + \{v, v\}) - q(C) \leq q(C) - 1 - q(C) = -1 < 0$. On the other hand, if v and the remaining white vertices are not adjacent, the number of connected components remains unchanged, resulting in $\Delta_{+\{v,v\}}q(C) = 0$. In summary, when v is a pink vertex,

$$\Delta_{+\{v,v\}}g(C) = \Delta_{+\{v,v\}}r(C) + \Delta_{+\{v,v\}}p(C) + \Delta_{+\{v,v\}}q(C) \leq -1 + 0 + 0 < 0.$$

This implies $g(C + \{v, v\}) < g(C)$.

Subcase 1.2. v is a white vertex.

then, $v \in S_0^{C^2}$, and $r_C(v) = k$. Similar to our approach with the pink vertex, let us analyze the three increments separately. For $\Delta_{+\{v,v\}}r(C)$,

$$\begin{aligned}\Delta_{+\{v,v\}}r(C) &= \sum_{u \in V} (r_{C+\{v,v\}}(u) - r_C(u)) \\ &= -r_C(v) - |N(v) \cap (\cup_{i=1}^{k-1} S_i^{C^2})| - |N(v) \cap S_0^{C^2}| \\ &= -k - |N(v) \cap (\cup_{i=0}^{k-1} S_i^{C^2})| \\ &\leq -k \\ &< 0,\end{aligned}$$

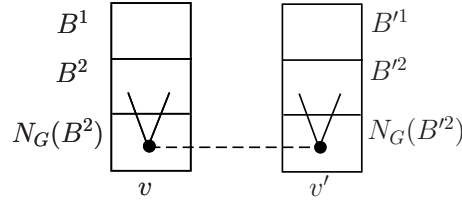


Figure 3

and hence $r(C + \{v, v\}) < r(C)$. For $\Delta_{+\{v, v\}}p(C)$, considering $v \in S_0^{C^2}$ implies that v has no neighbors in C^2 . If v^2 is added to C^2 , and v^2 has neighbors in C^1 , then $p(C + \{v, v\}) \leq p(C)$. On the other hand, if v^2 has no neighbors in C^1 , then $p(C + \{v, v\}) = p(C) + 1$, resulting in $\Delta_{+\{v, v\}}p(C) = 1$.

For $\Delta_{+\{v, v\}}q(C)$, if v^2 has neighbors in C^1 and v is adjacent to other white vertices, then $G\langle C \rangle$ reduces at least one connected component, and $\Delta_{+\{v, v\}}q(C) \leq -1$. If it is not adjacent to the remaining white vertices, the number of connected components of $G\langle C \rangle$ will decrease by at least one, leading to $\Delta_{+\{v, v\}}q(C) \leq -1$. If v^2 has no neighbor in C^1 and v is adjacent to other white vertices, then $G\langle C \rangle$ reduces at least one connected component, and $\Delta_{+\{v, v\}}q(C) \leq -1$. If it is not adjacent to the remaining white vertices, it reduces the number of connected components at the vertex v , resulting in $\Delta_{+\{v, v\}}q(C) = -1$. Thus, $\Delta_{+\{v, v\}}q(C) \leq -1$. Combining the case of $\Delta_{+\{v, v\}}p(C)$ and the case of $\Delta_{+\{v, v\}}q(C)$, we can conclude that $\Delta_{+\{v, v\}}p(C) + \Delta_{+\{v, v\}}q(C) \leq 0$ is true when v has neighbors in C^1 , and $\Delta_{+\{v, v\}}p(C) + \Delta_{+\{v, v\}}q(C) \leq 1 - 1 = 0$ is true when v has no neighbors in C^1 . In summary, when v is a white vertex,

$$\Delta_{+\{v, v\}}g(C) = \Delta_{+\{v, v\}}r(C) + \Delta_{+\{v, v\}}p(C) + \Delta_{+\{v, v\}}q(C) \leq -k + 0 + 0 < 0.$$

This implies $g(C + \{v, v\}) < g(C)$.

Case 2. C is an Roman k -dominating set, but $G[C]$ is not connected.

As C is a Roman k -dominating set, G contains only black, gray, and red vertices concerning C . Because $G[C]$ is not connected, there are at least two connected components: CP_C^1 and CP_C^2 in C .

If CP_C^1 and CP_C^2 have at least one common neighbor in $S_k^{C^2}$, then after placing v in C^1 , CP_C^1 and CP_C^2 merge into the same connected component. As a result, $p(C + \{v\}) < p(C)$. Moreover, since $q(C + \{v\}) = q(C)$, it follows that $g(C + \{v\}) < g(C)$.

If no connected components of $G[C]$ have a common neighbor, then $G\langle C \rangle$ is not connected. Every connected component of $G\langle C \rangle$ has the form $H_B = G[B] + N_G(B^2)$. This implies that H_B consists of a connected component of $G[C]$, induced by the vertex set B , with vertices in $N_G(B^2)$ (the red vertex set about B) attached to $G[B]$ (see Figure 3). Furthermore, the nearest components of $G\langle C \rangle$, say H_B and $H_{B'}$, are linked by an edge vv' with $v \in N_G(B^2)$ and $v' \in N_G(B'^2)$. Adding v^1 into C^1 , H_B and $H_{B'}$ merge into a larger connected subgraph of $G\langle C + \{v\} \rangle$. So, $q(C + \{v\}) < q(C)$, while $p(C + \{v\}) = p(C)$. Hence $g(C + \{v\}) < g(C)$. $\square$

Suppose Algorithm 1 adds multisets $X_1, X_2, \dots, X_g$ to C in sequence. Denote $C_i = X_1 + X_2 + \dots + X_i$ for $i = 1, \dots, g$, and let $C_0 = \emptyset$. Lemma 2 directly leads to the following corollary.

Corollary 3.1. For $i = 0, 1, \dots, g - 1$,

$$-\frac{\Delta_{+X_{i+1}}g(C_i)}{|X_{i+1}|} \geq \frac{1}{2}.$$

Proof. Since g is an integer-valued function, $g(C + \{v\}) < g(C)$ or $g(C + \{v, v\}) < g(C)$ implies $-\Delta_{+X}g(C) \geq 1$ for $X = \{v\}$ or $X = \{v, v\}$, indicating cost-effectiveness $-\frac{\Delta_{+X}g(C_i)}{|X|} \geq \frac{1}{2}$. The lemma is then deduced from the greedy strategy. $\square$

3.3 Theoretical Analysis

For the connected graph $G = (V, E)$, let f_{kR}^* be the minimum connected Roman k -dominating function of G . Denote $C^* = \{v \in V | f_{kR}^*(v) = 1 \text{ or } f_{kR}^*(v) = 2\}$. The lemma below provides a decomposition of C^* (where each part has at most one common vertex) and serves as a bridge between C^* and the output solution of Algorithm 1.

Lemma 3. For any $m \geq 2$, C^* can be decomposed into $Y_1, Y_2, \dots, Y_h$ for some $h \geq 1$, such that

- (a) $C^* = Y_1 \cup Y_2 \cup \dots \cup Y_h$;
- (b) For each $1 \leq i \leq h$, both $G[Y_i]$ and $G[Y_1 \cup Y_2 \cup \dots \cup Y_i]$ are connected;
- (c) $1 \leq f_{kR}^*(Y_i) \leq 2m - 1$ for $i = 1, \dots, h$, and at least $h - 1$ parts satisfy $f_{kR}^*(Y_i) \geq m + 1$;
- (d) $f_{kR}^*(Y_1) + f_{kR}^*(Y_2) + \dots + f_{kR}^*(Y_h) \leq f_{kR}^*(C^*) + 2(h - 1)$.

Proof. Since $G[C^*]$ is connected, there exists a spanning tree of $G[C^*]$. Let T be a spanning tree of $G[C^*]$, and choose an arbitrary vertex $v \in C^*$ as the root of T . For any vertex $x \in C^*$, denote by $T(x)$ the subtree of T rooted at x , and let $f_{kR}^*(T(x)) = \sum_{v \in V(T(x))} f_{kR}^*(v)$.

If $f_{kR}^*(T) \leq 2m - 1$, then set $h = 1$ and $Y_1 = C^*$; all three conditions are satisfied.

If $f_{kR}^*(T) \geq 2m$, then there exists a vertex x with $f_{kR}^*(T(x)) \geq m + 1$, and all of its children, denoted as $y_1, \dots, y_l$, satisfy $f_{kR}^*(T(y_i)) \leq m$. We will iteratively construct a connected vertex set Y and then proceed to analyze the decomposition of the remaining tree T' , ensuring that Y and T' share at most one common vertex.

If $f_{kR}^*(x) = 1$, and one child y_i has $f_{kR}^*(T(y_i)) = m$, then we let $Y = T(y_i) \cup \{x\}$ and $T' = T - T(y_i)$. In this case, $f_{kR}^*(Y) = m + 1$. If $f_{kR}^*(T(y_i)) \leq m - 1$, since $f_{kR}^*(T(x)) \geq m + 1$ implies $\sum_{i=1}^l f_{kR}^*(T(y_i)) \geq m$, there is an index j satisfying

$$f_{kR}^*(T(y_1)) + \dots + f_{kR}^*(T(y_j)) \leq m - 1$$

and

$$f_{kR}^*(T(y_1)) + \dots + f_{kR}^*(T(y_{j+1})) \geq m.$$

Let $Y = T(y_1) \cup \dots \cup T(y_{j+1}) \cup \{x\}$ and $T' = T - (T(y_1) \cup \dots \cup T(y_{j+1}))$. Note that $f_{kR}^*(Y) \geq m + 1$ and $f_{kR}^*(Y) \leq 2m - 1$.

Next, suppose $f_{kR}^*(x) = 2$. If one child y_i has $f_{kR}^*(T(y_i)) = m$, then we let $Y = T(y_i) \cup \{x\}$ and $T' = T - (T(y_i))$. In this case, $f_{kR}^*(Y) = m + 2$. If $f_{kR}^*(T(y_i)) \leq m - 2$ for any $i = 1, \dots, l$, similar to the above analysis, there exists an index j with $f_{kR}^*(T(y_1)) + \dots + f_{kR}^*(T(y_j)) \leq m - 2$

and $f_{kR}^*(T(y_1)) + \dots + f_{kR}^*(T(y_{j+1})) \geq m - 1$. Let $Y = T(y_1) \cup \dots \cup T(y_{j+1}) \cup \{x\}$ and $T' = T - (T(y_1) \cup \dots \cup T(y_{j+1}))$. In this case, $f_{kR}^*(Y) = m + 2$. The decomposition continues in the same manner until we obtain a set Y such that $f_{kR}^*(Y) \leq 2m - 1$.

Iteratively using such a construction method, we obtain a sequence of sets Y . If ordered in the reverse order of construction, then they satisfy conditions (b) and (c). Note that in the above construction, x is the only vertex common to Y and T' . So, every time a Y is constructed (except for the last Y when the remaining tree T' has $f_{kR}^*(T') \leq 2m - 1$), Y contains one vertex which will appear again in the later decompositions. Combining this with $f_{kR}^*(x) \leq 2$, condition (d) is satisfied. $\square$

Lemma 4. For any $0 \leq i \leq g - 1$ and $2 \leq j \leq h$,

$$-\Delta_{Y_j}g(C_i \cup C_{j-1}^*) \leq -\Delta_{+Y_j}g(C_i) + 2. \quad (5)$$

Proof. Note that

$$-\Delta_{Y_j}g(C_i \cup C_{j-1}^*) = -\Delta_{Y_j}r(C_i \cup C_{j-1}^*) - \Delta_{Y_j}p(C_i \cup C_{j-1}^*) - \Delta_{Y_j}q(C_i \cup C_{j-1}^*) = g(C_i \cup C_{j-1}^*) - g(C_i \cup C_j^*)$$

and

$$-\Delta_{+Y_j}g(C_i) = -\Delta_{+Y_j}r(C_i) - \Delta_{+Y_j}p(C_i) - \Delta_{+Y_j}q(C_i) = g(C_i) - g(C_i + Y_j).$$

We prove this lemma by proving the following claims.

Claim 1. $-\Delta_{Y_j}r(C_i \cup C_{j-1}^*) \leq -\Delta_{+Y_j}r(C_i)$.

Proof. Let us prove this claim through the bridge of $-\Delta_{Y_j}r(C_i)$. We first prove that $-\Delta_{Y_j}r(C_i) \leq -\Delta_{+Y_j}r(C_i)$, and then prove that $-\Delta_{Y_j}r(C_i \cup C_{j-1}^*) \leq -\Delta_{Y_j}r(C_i)$, so that $-\Delta_{Y_j}r(C_i \cup C_{j-1}^*) \leq -\Delta_{+Y_j}r(C_i)$. Observe that

$$\begin{aligned} -\Delta_{+Y_j}r(C_i) - (-\Delta_{Y_j}r(C_i)) &= (r(C_i) - r(C_i + Y_j)) - (r(C_i) - r(C_i \cup Y_j)) \\ &= r(C_i \cup Y_j) - r(C_i + Y_j) \\ &= \sum_{v \in V} (r_{C_i \cup Y_j}(v) - r_{C_i + Y_j}(v)), \end{aligned}$$

We consider each $v \in V$ to consider the value of each term in the sum.

If $v \in S_0^{C_i \cup Y_j}$, then the difference between $C_i \cup Y_j$ and $C_i + Y_j$ shows that if $v \in S_0^{C_i \cup Y_j}$, then $v \in \cup_{i=0}^k S_i^{C_i + Y_j}$, and hence

$$(r_{C_i \cup Y_j}(v) - r_{C_i + Y_j}(v)) = k - (k - i) = i \geq 0.$$

Suppose that $v \in S_i^{C_i \cup Y_j}$ ($i = 1, 2, \dots, k - 1$). If $v \in S_i^{C_i \cup Y_j}$ ($i = 1, 2, \dots, k - 1$), then $v \in \cup_{i=0}^k S_i^{C_i + Y_j}$, and hence

$$(r_{C_i \cup Y_j}(v) - r_{C_i + Y_j}(v)) \geq (k - i) - (k - i) = 0.$$

Suppose that $v \in S_k^{C_i \cup Y_j}$. If $v \in S_k^{C_i \cup Y_j}$, then $v \in S_k^{C_i + Y_j}$. Hence

$$(r_{C_i \cup Y_j}(v) - r_{C_i + Y_j}(v)) = 0 - 0 = 0.$$

From the above three cases, each term in the sum is greater than or equal to 0. Therefore, we have successfully proved that $-\Delta_{Y_j}r(C_i) \leq -\Delta_{+Y_j}r(C_i)$.

Now we prove $-\Delta_{Y_j}r(C_i \cup C_{j-1}^*) \leq -\Delta_{Y_j}r(C_i)$. We consider the relationship between $-\Delta_{Y_j}r(C_i \cup C_{j-1}^*)$ and $-\Delta_{Y_j}r(C_i)$ in terms of their definitions. Observe that

$$\begin{aligned} -\Delta_{Y_j}r(C_i) &= \sum_{v \in V} (r_{C_i}(v) - r_{C_i \cup Y_j}(v)) \\ &= (r_{C_i}(x_1) + \cdots + r_{C_i}(x_l)) + |N(Y_j^2) \cap (\cup_{i'=0}^{k-1} S_{i'}^{C_i^2})| \end{aligned}$$

and

$$\begin{aligned} -\Delta_{Y_j}r(C_i \cup C_{j-1}^*) &= \sum_{v \in V} (r_{C_i \cup C_{j-1}^*}(v) - r_{C_i \cup C_{j-1}^* \cup Y_j}(v)) \\ &= (r_{C_i \cup C_{j-1}^*}(y_1) + \cdots + r_{C_i \cup C_{j-1}^*}(y_{l'})) + |N(Y_j^2) \cap (\cup_{i'=0}^{k-1} S_{i'}^{(C_i \cup C_{j-1}^*)^2})|, \end{aligned}$$

where $Y_j \setminus C_i = \{x_1, \dots, x_l\}$, $Y_j \setminus (C_i \cup C_{j-1}^*) = \{y_1, \dots, y_{l'}\}$. Since $C_i \subseteq C_i \cup C_{j-1}^*$, then $l \geq l'$ and $r_{C_i}(x_l) \geq r_{C_i \cup C_{j-1}^*}(y_{l'})$. If any $v \in N(Y_j^2) \cap (\cup_{i'=0}^{k-1} S_{i'}^{(C_i \cup C_{j-1}^*)^2})$, then $v \in N(Y_j^2) \cap (\cup_{i'=0}^{k-1} S_{i'}^{C_i^2})$. Hence $-\Delta_{Y_j}r(C_i) \geq -\Delta_{Y_j}r(C_i \cup C_{j-1}^*)$. Claim 1 is available. $\square$

Claim 2. $-\Delta_{Y_j}p(C_i \cup C_{j-1}^*) \leq -\Delta_{+Y_j}p(C_i) + 1$.

Proof. Note that for any vertex set C and a connected vertex set Y , both $-\Delta_Y p(C)$ and $-\Delta_{+Y} p(C)$ equal the number of connected components of $G[C]$ that are adjacent to Y minus 1 (if a connected component of $G[C]$ has a common vertex with Y , then it is also regarded as adjacent to Y). By (b) of Lemma 3, $G[Y_j]$ is connected, and C_{j-1}^* induces a connected subgraph. So, compared with $G[C_i]$, subgraph $G[C_i \cup C_{j-1}^*]$ has at most one more connected component that is adjacent to Y_j , namely the one that contains C_{j-1}^* . Then, Claim 2 follows. $\square$

Claim 3. $-\Delta_{Y_j}q(C_i \cup C_{j-1}^*) \leq -\Delta_{+Y_j}q(C_i) + 1$.

Proof. For a vertex set C and a connected vertex set Y , we first consider the effect of $-\Delta_Y q(C)$ and $-\Delta_{+Y} q(C)$.

In the case $Y \cap C = \emptyset$, the effect of $-\Delta_Y q(C)$ and $-\Delta_{+Y} q(C)$ is the same. A connected component of $G\langle C \rangle$ with the form $H_B = G[B] + N_G(B^2)$ can be merged into $H_Y = G[Y] + N_G(Y^2)$ if and only if either B and Y are adjacent, or $N_G(B^2)$ and Y are adjacent, or B and $N_G(Y^2)$ are adjacent (having a common vertex is viewed as adjacent). A trivial component of $G\langle C \rangle$, which is a vertex v not dominated by C , can be merged into H_Y if and only if $v \in Y \cap N_G(Y^2)$. Call the above component H_B or the above trivial component v as an effective component of $G\langle C \rangle$ with respect to Y . Then, both $-\Delta_Y q(C)$ and $-\Delta_{+Y} q(C)$ equal the number of effective components with respect to Y .

In the case when $Y \cap C \neq \emptyset$, one big difference occurs when a vertex $v \in Y \cap C$ has its f_{kR} -value 1 in both C and Y . For $-\Delta_Y q(C)$, the contribution of $\cup\{v\}$ is the same as the above argument. While for $-\Delta_{+Y} q(C)$, v is moved from value 1 in C^1 to value 2 in $(C + Y)^2$. So v can dominate more vertices, and the consequence of $+\{v\}$ might merge more effective components than $\cup\{v\}$.

For the case $f_{kR}(v) = 1$ in C and $f_{kR}(v) = 2$ in Y , both the effect of $\cup\{v\}$ and $+\{v\}$ is to move v from value 1 to value 2. And for the case when $f_{kR}(v) = 2$ in C , we may ignore the effect of $\cup\{v\}$ and $+\{v\}$.

Recall that both $G[Y_j]$ and $G[C_{j-1}^*]$ are connected. Therefore, $C_i \cup C_{j-1}^*$ may have, at most, one additional effective component concerning Y_j compared to C_i . Specifically, this additional component would be the connected component containing C_{j-1}^* . Following the above argument, Claim 3 is established. $\square$

Theorem 3.1. *For any real number $\varepsilon > 0$, setting $m = \lceil \frac{2}{\varepsilon} + 1 \rceil$ in Algorithm 1, the approximation ratio for MinCRkDS is at most $(1 + \varepsilon)(2 + \ln((k + 1) + 2\Delta))$, where Δ is the maximum degree of the input graph.*

Proof. Suppose Algorithm 1 selects vertex sets $X_1, \dots, X_g$ in sequence. Define $C_i = X_1 + X_2 + \dots + X_i$ for $i = 1, \dots, g$, and set $C_0 = \emptyset$. It is important to note that the output of the algorithm is C_g . Consider f_{kR}^* as a minimum weight CRkDS function, and denote $opt = \sum_{v \in V} f_{kR}^*(v)$ as the optimal value. For $i = 0, 1, 2$, define $V_{f_{kR}^*}^i = \{v \in V \mid f_{kR}^*(v) = i\}$ and $C^* = V_{f_{kR}^*}^1 \cup V_{f_{kR}^*}^2$. Decompose C^* into $Y_1 \cup Y_2 \cup \dots \cup Y_h$ following the process outlined in Lemma 3. Define $C_j^* = Y_1 \cup Y_2 \cup \dots \cup Y_j$ for each $j = 1, \dots, h$, and set $C_0^* = \emptyset$.

Regarding the selection of X in line 3 of Algorithm 1, it is essential to recall that $|X|$ denotes the cardinality of the multi-set X . If we consider each $f_{kR}^*(Y_j)$ as $|Y_j|$ when treating Y_j as a multi-set meaning each $v \in Y_j$ has f_{kR}^* copies in Y_j then the condition $f_{kR}^*(Y_j) \leq 2m - 1$ (condition (ii) of Lemma 3) implies that Y_j is a valid candidate for X . The greedy selection of X ensures that:

$$\frac{-\Delta_{+X_{i+1}}g(C_i)}{|X_{i+1}|} \geq \frac{-\Delta_{+Y_j}g(C_i)}{|Y_j|} \quad (\forall 0 \leq i \leq g-1, 1 \leq j \leq h). \quad (6)$$

Combining this with Lemma 4, condition (iii) of Lemma 3, implies

$$\begin{aligned} \frac{-\Delta_{+X_{i+1}}g(C_i)}{|X_{i+1}|} &\geq \frac{-\sum_{j=1}^h \Delta_{+Y_j}g(C_i)}{\sum_{j=1}^h |Y_j|} \\ &\geq \frac{-2(h-1) - \sum_{j=1}^h \Delta_{Y_j}g(C_i \cup C_{j-1}^*)}{\sum_{j=1}^h |Y_j|} \\ &\geq \frac{-2(h-1) - (g(C_i \cup C_h^*) - g(C_i))}{opt + 2(h-1)} \\ &= \frac{-2(h-1) - (2 - g(C_i))}{opt + 2(h-1)} \\ &= \frac{g(C_i) - 2h}{opt + 2(h-1)}, \end{aligned} \quad (7)$$

where the first equality uses the observation that $C_i \cup C_h^* \supseteq C^*$ is also a connected Roman k -dominating set of G , and thus $g(C_i \cup C_h^*) = 2$ by Lemma 1.

Denote $a_i = g(C_i) - 2h$. Note that a_i is monotone decreasing with respect to i , and $a_0 = (k+1)|V| - 2h$. Inequality (7) leads to

$$\frac{a_i - a_{i+1}}{|X_{i+1}|} \geq \frac{a_i}{opt + 2(h-1)}, \quad (8)$$

and thus

$$a_{i+1} \leq a_i \left(1 - \frac{|X_{i+1}|}{opt + 2(h-1)} \right) \leq a_i e^{-\frac{|X_{i+1}|}{opt + 2(h-1)}}, \quad (9)$$

where the second inequality uses the fact $1 + x \leq e^x$ for any real number x . Recursively using inequality (9), we have

$$a_{i+1} \leq a_0 e^{-\frac{|X_{i+1}| + |X_i| + \dots + |X_1|}{opt + 2(h-1)}}. \quad (10)$$

Case 1. $|V| \geq \frac{1}{k+1}(opt + 2h)$.

In this case, $a_0 \geq opt$ and $a_g = 2 - 2h < opt$. Hence there exists an index $i_0 \in \{0, 1, \dots, g-1\}$ such that

$$a_{i_0} \geq opt > a_{i_0+1}.$$

Let $b = a_{i_0} - opt$ and $b' = opt - a_{i_0+1}$. Write $|X_{i_0+1}| = d + d'$ such that

$$\frac{b}{d} = \frac{b'}{d'} = \frac{a_{i_0} - a_{i_0+1}}{|X_{i_0+1}|}. \quad (11)$$

Note that for the computed CRkDS function f_{kR} , its value $\sum_{v \in V} f_{kR}(v) = |C_g|$ can be divided into two parts: $\sum_{i=1}^{i_0} |X_i| + d$ and $d' + \sum_{i=i_0+2}^g |X_i|$.

First, we prove

$$\sum_{i=1}^{i_0} |X_i| + d \leq (opt + 2(h-1)) \ln \frac{(k+1)|V| - 2h}{opt}. \quad (12)$$

Combining (8) and (11), we have

$$\frac{b}{d} = \frac{a_{i_0} - opt}{d} \geq \frac{a_{i_0}}{opt + 2(h-1)},$$

resulting in

$$opt \leq a_{i_0} \left(1 - \frac{d}{opt + 2(h-1)} \right) \leq a_{i_0} e^{-\frac{d}{opt + 2(h-1)}}.$$

Combining this with (10),

$$opt \leq a_0 e^{-\frac{d + |X_{i_0}| + \dots + |X_1|}{opt + 2(h-1)}} = ((k+1)|V| - 2h) e^{-\frac{d + |X_{i_0}| + \dots + |X_1|}{opt + 2(h-1)}}.$$

By a rearrangement, inequality (12) follows.

Next, we show

$$d' + \sum_{i=i_0+2}^g |X_i| \leq 2(opt + 2(h-1)). \quad (13)$$

By the Corollary 3.1, $\frac{-\Delta_{+X_{i+1}}g(C_i)}{|X_{i+1}|} \geq \frac{1}{2}$, which leads to $|X_{i+1}| \leq 2(-\Delta_{+X_{i+1}}g(C_i))$. By (11) and the observation $\frac{a_{i_0}-a_{i_0+1}}{|X_{i_0+1}|} = \frac{-\Delta_{+X_{i_0+1}}g(C_{i_0})}{|X_{i_0+1}|} \geq \frac{1}{2}$, we have $d' \leq 2b'$. Hence

$$\begin{aligned} d' + \sum_{i=i_0+2}^g |X_i| &\leq 2(b' - \sum_{i=i_0+2}^g \Delta_{+X_i}g(C_{i-1})) \\ &= 2(b' + g(C_{i_0+1}) - g(C_g)) \\ &= 2(opt - a_{i_0+1} + g(C_{i_0+1}) - 2) \\ &= 2(opt + 2(h-1)). \end{aligned}$$

This proves inequality (13).

Combining inequalities (12) and (13), we have

$$\sum_{v \in V} f_{kR}(v) = \sum_{i=1}^g |X_i| \leq (opt + 2(h-1)) \left(2 + \ln \frac{(k+1)|V| - 2h}{opt} \right). \quad (14)$$

By Lemma 3,

$$(h-1)(m+1) \leq f_{kR}^*(Y_1) + \dots + f_{kR}^*(Y_h) \leq opt + 2(h-1).$$

It follows that

$$(h-1) \leq \frac{opt}{m-1}. \quad (15)$$

Next, we show $\frac{(k+1)|V|-2h}{opt} \leq ((k+1) + 2\Delta)(1 + \frac{2}{m-1})$ by the following claim.

Claim 1. $\frac{(k+1)|V|-2h}{opt} \leq ((k+1) + 2\Delta)(1 + \frac{2}{m-1})$.

Proof. Taking $i = 1$ in inequality (7), we have

$$\frac{a_0}{opt + 2(h-1)} \leq \frac{-\Delta_{+X_1}g(C_0)}{|X_1|}.$$

We discuss the value of $\frac{-\Delta_{+X_1}g(C_0)}{|X_1|}$. Using

$$\begin{aligned} -\Delta_{+X_1}g(C_0) &= g(\emptyset) - g(\emptyset + X_1) \\ &= g(\emptyset) - (r(+X_1) + p(+X_1) + q(+X_1)) \\ &= (k+1)|V| - (r(+X_1) + p(+X_1) + q(+X_1)), \end{aligned}$$

next, let us delve into the corresponding values of the three aforementioned functions. It is noteworthy that:

$$\begin{aligned} r(+X_1) &= \sum_{v \in V} r_{\{+X_1\}}(v) \\ &= (k+1)|N(X_1^2) \setminus X_1^1| + k(|V| - |X_1| - |N(X_1^2) \setminus X_1^1|) \\ &= k|V| - k|X_1| - |N(X_1^2) \setminus X_1^1| \end{aligned}$$

and

$p(+X_1)$ = the number of connected components of the induced subgraph $G[X_1]$.

Here, we designate the number of connected components of the induced subgraph $G[X_1]$ as n_1 (where $n_1 \geq 1$), and accordingly, $p(+X_1) = n_1$. By the definition of $q(\cdot)$, we have

$$\begin{aligned} q(+X_1) &= \text{the number of connected components of the induced subgraph } G\langle X_1 \rangle \\ &= |V| - |X_1| - |N(X_1^2) \setminus X_1^1| + n_2, \end{aligned}$$

where n_2 ($n_2 \geq 1$) is the number of connected components of the induced subgraph $G[X_1 \cup (N(X_1^2) \setminus X_1^1)]$. Then

$$\begin{aligned} -\Delta_{+X_1}g(C_0) &= (k+1)|V| - (r(+X_1) + p(+X_1) + q(+X_1)) \\ &= (k+1)|V| - (k|V| - k|X_1| - |N(X_1^2) \setminus X_1^1|) - (n_1) \\ &\quad - (|V| - |X_1| - |N(X_1^2) \setminus X_1^1| + n_2) \\ &= (k+1)|X_1| + 2|N(X_1^2) \setminus X_1^1| - n_1 - n_2 \\ &\leq (k+1)|X_1| + 2\Delta|X_1| - 2 \\ &= (k+1+2\Delta)|X_1| - 2. \end{aligned}$$

Hence

$$\begin{aligned} \frac{(k+1)|V| - 2h}{opt + 2(h-1)} &= \frac{a_0}{opt + 2(h-1)} \leq \frac{-\Delta_{+X_1}g(\emptyset)}{|X_1|} \\ &\leq \frac{|X_1|((k+1) + 2\Delta) - 2}{|X_1|} \\ &\leq (k+1) + 2\Delta. \end{aligned}$$

Combine (15) and reformulate the inequality; thus, the claim is proven. $\square$

Substituting (15) and Claim 1 into (14), we have

$$\begin{aligned} \sum_{v \in V} f_{kR}(v) &= \sum_{i=1}^g |X_i| \leq (opt + 2(h-1)) \left(2 + \ln \frac{(k+1)|V| - 2h}{opt} \right) \\ &\leq (opt + 2(h-1)) \left(2 + \ln(k+1+2\Delta) \left(1 + \frac{2}{m-1} \right) \right) \\ &= opt \left(1 + \frac{2(h-1)}{opt} \right) \left(2 + \ln(k+1+2\Delta) \left(1 + \frac{2}{m-1} \right) \right) \\ &\leq opt \left(1 + \frac{2}{m-1} \right) (2 + \ln(k+1+2\Delta) \left(1 + \frac{2}{m-1} \right)) \\ &\leq opt(1 + \varepsilon)(2 + \ln(k+1+2\Delta)) \end{aligned}$$

by setting $\frac{1}{m-1} < \varepsilon$.

Case 2. $|V| < \frac{1}{k+1}(opt + 2h)$.

Using Corollary 3.1, we have

$$\begin{aligned}\sum_{v \in V} f_{kR}(v) &= \sum_{i=1}^g |X_i| \leq 2 \sum_{i=1}^g -\Delta_{+X_{i+1}} g(C_i) = 2(g(C_0) - g(C_g)) \\ &= 2((k+1)|V| - 2) < 2(opt + 2(h-1)).\end{aligned}$$

Then by (15) and the choice of k ,

$$\sum_{v \in V} f_{kR}(v) < opt \cdot 2(1 + \varepsilon).$$

Hence, the desired approximation ratio is proved. $\square$

4 Conclusions

In this paper, we employed a greedy strategy to formulate a $(1+\varepsilon)(2+\ln(k+1+2\Delta))$ -approximation algorithm for MinCR k DS. Addressing this problem involves overcoming four challenges: handling the non-submodular potential function, dynamically managing the vertex costs, grappling with an inequality crucial for analyzing the approximation ratio, and establishing a connection between the optimal solution and the approximate solution. There are two open questions here, one is how can we enhance the algorithm to achieve a better approximation ratio? Second, is it possible to get good inapproximability bounds?

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